

The image shows a Visual Studio Code editor window with a Python script named `DecisionTree_2.py` open. The script uses `pandas` to read a CSV file, `sklearn.tree` to fit a `DecisionTreeClassifier`, and `matplotlib.pyplot` to plot the decision tree. The data is loaded from `Data_DecisionTree.csv`. The model is trained on a dataset with features `tempMode`, `AQ`, `USS`, `CS`, `VOC`, `RP`, `IP`, and `Temperature`. The input data for prediction is `[[4, 5, 3, 6, 1, 45, 5, 1]]`. The model predicts `1`, which corresponds to 'Failure' in the `class_names` list. The terminal output shows the prediction result and a warning from `sklearn` about invalid feature names.

```
1 import pandas as pd
2 from sklearn.tree import DecisionTreeClassifier
3 from sklearn.tree import plot_tree
4 import matplotlib.pyplot as plt
5
6 data = pd.read_csv('Data_DecisionTree.csv')
7 X = data[['tempMode', 'AQ', 'USS', 'CS', 'VOC', 'RP', 'IP', 'Temperature']]
8 y = data['fail']
9
10 model = DecisionTreeClassifier(criterion='entropy')
11 model.fit(X, y)
12 input_data = [[4, 5, 3, 6, 1, 45, 5, 1]]
13
14
15 prediction = model.predict(input_data)
16 print("Prediction Result:", prediction[0])
17 if prediction[0] == 1:
18     print("The machine will fail.")
19 else:
20     print("The machine will not fail.")
21
22 plt.figure(figsize=(18, 10))
23 plot_tree(model, feature_names=['tempMode', 'AQ', 'USS', 'CS', 'VOC', 'RP', 'IP', 'Temperature'], class_names=['No Failure', 'Failure'], filled=True)
24 plt.title("Decision Tree Classifier")
25 plt.show()
```

Terminal Output:

```
PS D:\coding> & C:\Users\alphi\AppData\Local\Programs\Python\Python314\python.exe "d:/coding/New folder/DecisionTree_2.py"
C:\Users\alphi\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\utils\validation.py:2691: UserWarning: X does not have valid feature names, but DecisionTreeClassifier was fitted with feature names
  warnings.warn(
Prediction Result: 1
The machine will fail.
[]
```

VS Code Interface Details:

- Explorer: Shows the project structure with files like `aj.py`, `Assignment 2_PowerO...`, `cancer_data.csv`, `Cancer_predictor.py`, `Cancer.py`, `Data_DecisionTree.csv`, `data.csv`, `Decision_Tree.csv`, `Decision_Tree.py`, `KNN_Dataset.csv`, `KNN_FuelConsumptio...`, `LoadExtension_ALPHI...`, `LoadExtension.py`, `machine_failure.csv`, `Machine_Failure.py`, `ml_program.py`, `model.pkl`, `power_data.csv`, and `sasfds.py`.
- Terminal: Shows the command prompt and the execution of the Python script.
- Status Bar: Shows the current line and column (Ln 17, Col 23), encoding (UTF-8), and the Python interpreter (Python 3.14).

